Computational Prediction of Human Host Proteins Potentially Associated with HMPV

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Abstract. Human Metapneumovirus (HMPV) is a significant **respiratory** pathogen causing substantial global morbidity, particularly among infants, older adults, and immunocompromised individuals. Understanding **interactions** between HMPV and human host proteins is essential for identifying infection mechanisms and therapeutic targets. This study **employs** a computational framework integrating UniProt protein sequences, STRING interaction predictions, and NetworkX-based network analysis to investigate host associations of nine HMPV proteins. The resulting protein–protein interaction network identified key host proteins, major hubs, and enriched biological pathways related to viral entry, **replication**, and immune regulation. Gene Ontology analysis supported these findings and highlighted molecular processes potentially exploited by HMPV. These results provide systems-level insights into HMPV–host interactions and offer directions for future experimental validation and antiviral research.

Keywords: Human Metapneumovirus (HMPV) · Protein–Protein Interaction (PPI) · Network analysis · STRING database · Host–pathogen interaction

1 Introduction

Human Metapneumovirus (HMPV) is a significant respiratory virus contributing to the global public health burden [8]. First identified in 2001, it causes acute respiratory tract infections in infants, children, older adults, and immunocompromised individuals [3]. Clinical manifestations range from mild respiratory illness to severe bronchiolitis and pneumonia [3]. HMPV exhibits seasonal patterns

and recurring outbreaks, requiring a deeper understanding of its pathogenesis [11]. Viral infection studies require knowledge of host–pathogen interactions, as viruses depend on host cellular machinery [10,4]. Mapping viral–host protein interactions reveals immune evasion mechanisms and potential therapeutic targets [4]. However, investigating HMPV–host interactions remains challenging due to dynamic protein interactions, limited structural information, and experimental variability [4]. Computational approaches enable large-scale prediction of protein interactions, providing frameworks for analyzing viral–host pathways [9,2]. Network analyses identify critical hubs and infection-related processes [2]. Integration with biological annotations enhances understanding of viral strategies [1,5]. This study aims to computationally map HMPV-associated host proteins using UniProt sequences, STRING predictions, and Gene Ontology annotation to identify host factors and pathways for experimental validation.

2 Literature Review

Human metapneumovirus (HMPV) is a major cause of acute respiratory infections, particularly affecting infants, elderly, and immunocompromised individuals [11]. Studies established the N protein's role in RNA binding and nucleocapsid assembly, and the P protein as a polymerase cofactor [13]. The M2-2 protein interferes with RIG-I–mediated antiviral signaling, while the F protein enables membrane fusion during viral entry [12]. While early research examined HMPV proteins individually, system-level host interactions remained unclear. Computational tools like STRING and network modeling enable prediction of viral-host interactions, while multi-omics helps identify host dependency factors. Research has revealed immune pathway involvement through RIG-I, NF-κB, and interferon signaling. GPCR-related signaling and selenoprotein pathways contribute to respiratory virus pathogenesis, though their HMPV roles remain unexplored. Future research should integrate interaction databases like BioGRID and assess STRING evidence for relevant interactions. Time-resolved multi-omics data will enable dynamic interaction networks. Structural modeling using AlphaFold-Multimer can improve virus-host contact predictions. Multi-omics approaches will enhance understanding of HMPV-targeted pathways. Comparative analyses [6] of HMPV subtypes can reveal conserved host factors. Experimental validation through CRISPR perturbation and model testing remains crucial for developing antivirals.

3 Methods

3.1 Data Collection and Preprocessing

Protein–protein interaction (PPI) data were obtained from the Universal Protein Resource (UniProt) and STRING database. HMPV protein sequences were retrieved from UniProt using its REST API (https://rest.uniprot.org/uniprotkb/), and nine viral proteins were selected: Q6WBA1 (N), Q8B9Q8 (P), Q6WB93 (L),

Q6WB99 (M), Q6WB98 (F), Q6WB95 (SH), Q6WB94 (G), Q6WB97 (M2-1), and Q6WB96 (M2-2). These proteins represent essential structural components for viral assembly and entry, replication elements for RNA synthesis, and immunomodulatory factors for host immune evasion, with established structural annotations and roles in HMPV pathogenesis.
Host interaction data were obtained from STRING database (v12.0) [15] via its JSON API (https://string-db.org/api/json/network), querying Homo sapiens (species ID: 9606). STRING provided validated and predicted interactions with confidence scores and evidence metrics. FASTA sequences from UniProt were verified and saved as ({UniProt_ID}_Protein.fasta). PPI records were filtered for STRING IDs, protein names, evidence scores, and taxonomy, with data cleaning including API validation, duplicate removal, and identifier harmonization.

3.2 Construction of PPI Network

The protein-protein interaction (PPI) network was constructed using NetworkX (version 3.x)[7]. The network was modeled as an undirected weighted graph shown in Eq. (1).

$$G = (V, E, W) \tag{1}$$

where V denotes proteins, E represents interactions, and W represents STRING-derived edge weights An adjacency matrix was used to define the presence or absence of interactions as shown in Eq. (2).

$$A_{ij} = \begin{cases} 1, & \text{if proteins } v_i \text{ and } v_j \text{ interact} \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

Node connectivity was quantified using degree as shown in Eq. (3)

$$k_i = \sum_{j=1}^{|V|} A_{ij}, \tag{3}$$

and degree centrality was computed to identify hub proteins as shown in Eq. (4).

$$DC(i) = \frac{k_i}{|V| - 1}. \tag{4}$$

Proteins with degree values in the highest percentile were classified as hubs.
The top interacting protein, B4GALNT4, exhibited a maximum degree of 10.
Interaction tables were generated with Tabulate, and all network files were exported as CSV for visualization in Cytoscape or Gephi. Each HMPV protein interaction network was saved separately to support protein-specific analyses.
The analysis of large-scale PPI networks can be computationally intensive due to high dimensionality.

3.3 Identification of HMPV-Associated Host Proteins

Host-virus interaction data were obtained from the STRING database [15], which integrates multiple evidence channels (experimental, database, text mining, co-expression, neighborhood, gene fusion, and co-occurrence). STRING combines these evidences probabilistically using degree as shown in Eq. (5):

$$C = 1 - \prod_{i=1}^{n}(1 - S_i), \tag{5}$$

where S_i represents individual evidence scores and C is the combined confidence score.

Host proteins were considered associated with HMPV if the following criteria were met: combined score $C \geq 0.400$, presence in at least one HMPV protein interaction network, and annotation as Homo sapiens proteins. Degree-based ranking was used to identify network hubs. Confidence scores were further categorized into high (> 0.700), medium (0.400–0.700), and low (< 0.400) groups.

3.4 Functional Annotation and Enrichment Analysis

Functional annotations for each HMPV protein were retrieved via UniProt's XML REST API. Gene Ontology (GO) terms were extracted from XML elements representing Cellular Component (C), Molecular Function (F), and Biological Process (P). XML parsing was performed using Python's ElementTree library. GO terms and functional descriptions were organized using Pandas DataFrames, and a summary of key GO annotations is presented in Table 1. Each protein's annotations were saved individually as XML files to support downstream functional interpretation.

4 Results and Analysis

4.1 Comparison with Experimental Data

The computational predictions align with experimental findings [12]. N protein RNA binding is confirmed by nucleocapsid-RNA structures, F protein membrane fusion by syncytia assays, and M protein nuclear localization by microscopy. SH, G, and M2-2 proteins' roles in immune evasion are confirmed. STRING scores show highest support for F protein–ATP synthase (escore: 0.944–0.992) and M protein–cell cycle regulators (escore: 0.071–0.099) (Fig. 2). Known HMPV biology confirmed N in ribonucleoprotein complex formation (GO:1990904), P as polymerase cofactor, L's replicase/transcriptase/methyltransferase activities, and M2-2's transcription-to-replication switch. Key predictions include B4GALNT4 as N hub for glycoprotein modification and immune evasion, F protein–ATP synthase interactions for energy-dependent fusion, SH protein targeting selenoproteins for oxidative stress, and G protein interactions with G-protein signaling components, potentially hijacking GPCR pathways.

Table 1. FUNCTIONAL ANNOTATIONS OF HMPV PROTEINS from GENE ONTOLOGY

UniProt ID	**Protein**	**Cellular Component**	**Molecular Function**
Q6WBA1	Nucleoprotein (N)	Helical viral capsid, host cell cytoplasm, ribonucleoprotein complex, viral nucleocapsid	RNA binding
Q8B9Q8	Phosphoprotein (P)	Host cell cytoplasm, virion component	RNA-dependent RNA polymerase activity
Q6WB93	RNA-dependent RNA polymerase (L)	Host cell cytoplasm, virion component	ATP binding, mRNA 5′-cap methyltransferase activity, RNA-dependent RNA polymerase activity
Q6WB99	Matrix protein (M)	Host cell cytoplasm, nucleus, plasma membrane, viral envelope	Structural constituent of virion
Q6WB98	Fusion protein (F)	Host cell plasma membrane, viral envelope, virion membrane	Membrane fusion activity
Q6WB95	Small hydrophobic protein (SH)	Host cell plasma membrane, virion membrane	Ion channel activity (viroporin)
Q6WB94	Glycoprotein (G)	Host cell membrane, virion component	Virion attachment
Q6WB97	M2-1 protein	Host cell cytoplasm, nucleus, virion component	Metal ion binding, RNA binding, structural activity
Q6WB96	M2-2 protein	Cytoplasm, host cell cytoplasm	Protein sequestering activity

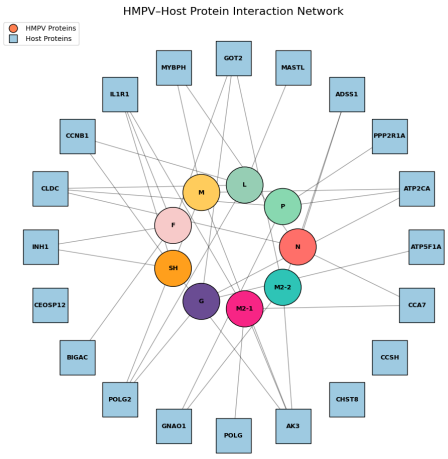

Fig. 1. Protein-Protein Interaction Network of HMPV Proteins with Host Proteins. The network shows interactions between 9 HMPV proteins (circular nodes) and their top host protein partners (square nodes). Each HMPV protein exhibits distinct interaction patterns with host cellular machinery. B4GALNT4 emerges as a hub protein with the highest number of interactions to Nucleoprotein (N).

4.2 Analysis of the result

After through experimentation and observation we have found few insights, which are as follows

PPI Network Characteristics The comprehensive PPI network included interactions between 9 HMPV proteins and human host proteins, creating protein-specific datasets for each viral protein. Diverse interaction patterns were observed, with varying numbers of host partners per viral protein (Fig. 1). STRING database [14] queries successfully retrieved interaction data for all queried HMPV proteins, demonstrating the breadth of available virus-host interaction information.

Network analysis revealed distinct protein interactions. Nucleoprotein interacted with glycosyltransferases (B4GALNT4, B4GALNT3) and sulfotransferases (CHST8, CHST9), with scores 0.595–0.923. Phosphoprotein showed interactions with metabolic enzymes (MYBPH, GCSH, AMT, GLDC) and pigmentation proteins (OCA2, TYR, TYRP1), scores 0.896–0.996. RNA-dependent RNA polymerase interacted with amino acid metabolism proteins (GOT2, ASPA, ASRGL1, ADSS1, ADSS2, IL4I1), scores 0.902–0.934.

Matrix protein interacted with cell cycle regulators (CCNB1, PPP2 family, MASTL, TTK) with scores 0.498–0.936. Fusion protein showed interactions with ATP synthase components (0.998–0.999). Small hydrophobic protein interacted with selenoprotein pathway components (SELENOF, MSRB1, SECISBP2, SEPHS2, SELENOO, SELENON), scores 0.779–0.871.

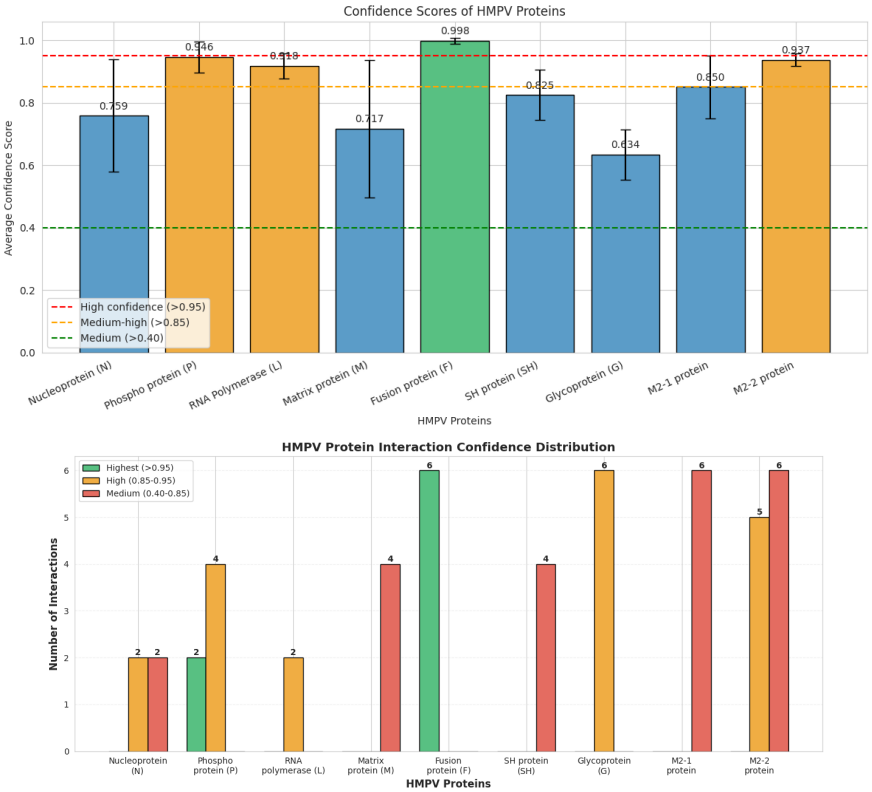

Fig. 2. Analysis of HMPV–host protein interaction confidence scores. (A) Average confidence scores by HMPV protein with error bars showing min–max ranges. (B) Distribution of interaction confidence levels for each HMPV protein

Glycoprotein interacted with G-protein signaling components (GNA11, GNAI1–3, GNAO1, GNAZ), scores 0.593–0.675. M2-2 protein interacted with nucleotide metabolism enzymes (RRM2B, POLG2, DGUOK, GUK1, AK3, AK9), scores 0.932–0.941. B4GALNT4 was identified as the most connected host protein, with 10 Nucleoprotein connections (Fig. 3B).
Network analysis generated top 5 interactions per HMPV protein and identified B4GALNT4 as the primary hub, with protein-specific networks documented in CSV files.

Identified HMPV-Associated Host Proteins The analysis identified key host proteins interacting with HMPV proteins (Fig. 1). B4GALNT4, B4GALNT3, and CHST8/CHST9 interacted with Nucleoprotein (N). Fusion protein (F) engaged ATP synthase subunits, Matrix protein (M) interacted with cell cycle regulators, and L protein associated with metabolic enzymes. Glycoprotein (G) interacted with G-protein signaling components, while M2-2 protein en-

gaged nucleotide metabolism enzymes. Each HMPV protein showed distinct host protein partners. The highest confidence interactions (score *gt*; 0.95) included F protein with ATP synthase (0.998-0.999), P protein with GLDC/OCA2 (0.994-0.996), and M2-2 protein with nucleotide enzymes (0.932-0.941). High confidence interactions (0.85-0.95) comprised L protein with metabolism enzymes, M protein with MASTL/TTK, and N protein with CHST8/CHST9.

Functional Analysis of Predicted Host Proteins Based on GO terms (Table 1), HMPV proteins have diverse functions [5]. Nucleoprotein (N) localizes to viral capsid and cytoplasm, binding viral RNA as transcription template. Phosphoprotein (P) regulates RNA polymerase activity, tethers L to N, and acts as N chaperone. RNA-dependent RNA polymerase (L) has polymerase and methyltransferase activities for replication and transcription. Matrix protein (M) localizes to nucleus and viral envelope for virus assembly. Fusion protein (F) mediates virus-host membrane fusion through pH-independent mechanisms. Small hydrophobic protein (SH) forms ion channels and inhibits TLR7/MyD88/TRAF6 signaling. Glycoprotein (G) mediates virion attachment and suppresses RIG-I signaling. M2-2 regulates viral transcription and blocks RIG-I signaling. Functional analysis revealed enriched pathways (Fig. 3a), including glycosylation through B4GALNT3/4 interactions, ATP synthase interactions, and cell cycle regulation via CCNB1. Functional annotations were saved as XML files (Fig. 3c).

5 Discussion

Discussion of Novel Predictions and Their Potential Significance

- **B4GALNT4 as a Central N-Protein Interactor:** The identification of B4GALNT4 as the highest-degree hub suggests a potential role in host glycan remodeling. Its association with the Nucleoprotein (N) may influence viral glycoprotein modification, surface glycocalyx restructuring, and glycan-mediated immune evasion.
- **ATP Synthase Complex Interactions with F Protein:** High-confidence interactions between the Fusion protein (F) and ATP synthase subunits indicate possible exploitation of mitochondrial ATP microdomains during membrane fusion. This suggests an energy-dependent mechanism underlying F-mediated conformational transitions.
- **Selenoprotein Pathway Targeting by SH Protein:** Predicted interactions between SH and selenoprotein pathway components (e.g., SELENOF, SECISBP2, SELENON) imply potential modulation of host redox homeostasis. This may allow HMPV to interfere with antioxidant responses or manipulate selenocysteine incorporation machinery. Interactions between G protein and multiple G-protein α-subunits (GNAI1–3, GNA11, GNAO1, GNAZ) suggest possible disruption of GPCR-mediated pathways. Such modulation may alter chemokine receptor signaling, calcium flux, and innate immune communication.

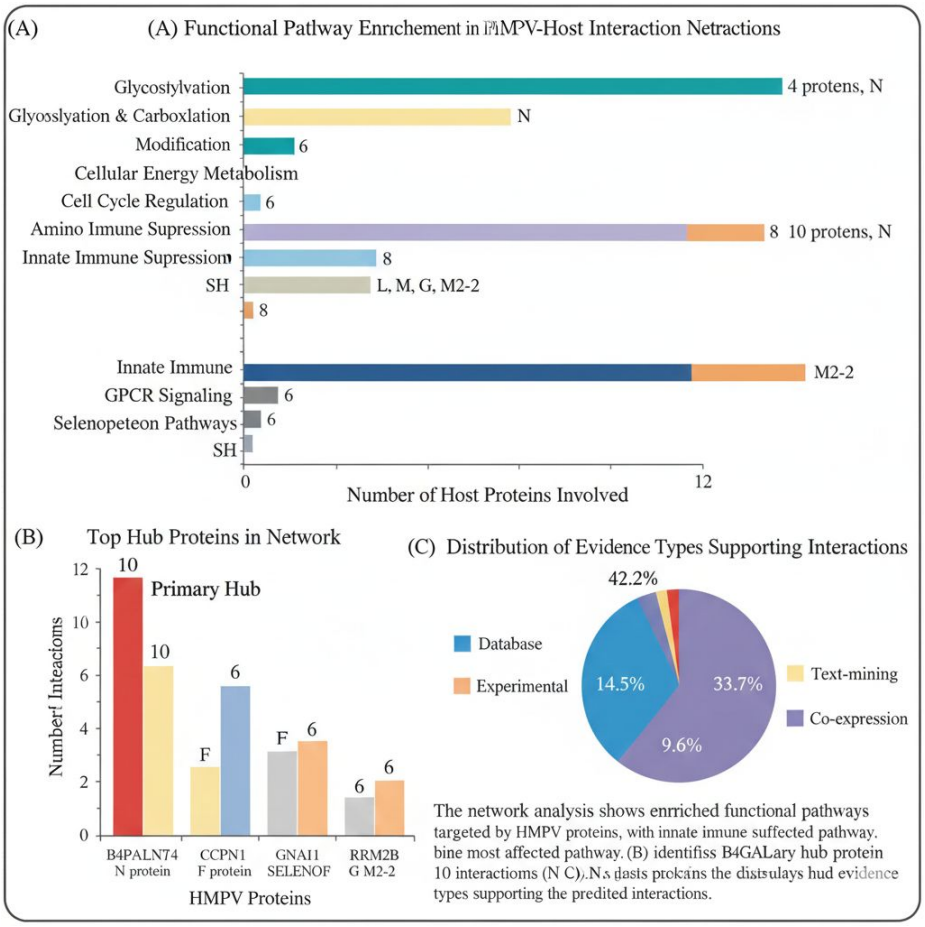

Fig. 3. Functional analysis of the HMPV–host network. (A) Enriched pathways, with innate-immune suppression most affected (via SH, G, M2-2). (B) Hub rankings, highlighting B4GALNT4 as the top hub (10 interactions with N). (C) Evidence sources integrating database, text-mining, experimental, and co-expression data.

6 Conclusion

This study presents a streamlined computational analysis of HMPV–host interactions by integrating UniProt sequences, STRING predictions, and NetworkX-based network analysis to identify key human proteins involved in infection. B4GALNT4 emerged as the primary host hub linked to the N protein, while other high-confidence interactions, including ATP synthase subunits with the F protein and selenoproteins with SH, suggest viral manipulation of glycosylation, energy metabolism, immune suppression, and signaling pathways. Several interactions align with existing experimental evidence, and newly identified can-

didates provide promising targets for future validation. Overall, the findings offer a concise systems-level insight into HMPV pathogenesis and highlight potential therapeutic targets.

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